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A late Neoproterozoic paleomagnetic pole for the Congo craton: Tectonic setting, paleomagnetism and geochronology of the Nola dike swarm (Central African Republic)

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ABSTRACT

New structural, geochronological and paleomagnetic data were obtained on dolerite dikes of the Nola region (Central African Republic) at the northern border of the Congo craton. In this region, metavolcanic successions were thrust southward onto the craton during the Panafrican orogenic events. Our structural data reveal at least two structural klippes south of the present-day limits of the Panafrican nappe suggesting that it has once covered the whole Nola region, promoting the pervasive hydrothermal greenschist metamorphism observed in the underlying cratonic basement and also in the intrusive dolerite dikes. Paleomagnetic measurements revealed a stable dual-polarity low-inclination magnetization component in nine dikes (47 samples), carried by pyrrhotite and magnetite. This component corresponds to a paleopole at 304.8°E and 61.8°S ($dp = 5.4$, $dm = 10.7$) graded at $Q = 6$. Both metamorphism and magnetic resetting were dated by the $^{40}Ar/^{39}Ar$ method on amphibole grains separated from the dikes at 571 ± 6 Ma. The Nola pole is the first well-dated paleomagnetic pole for the Congo craton between 580 and 550 Ma. It marks a sudden change in direction of the Congo craton apparent polar wander path at the waning stages of the Panafrican orogenic events.

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1. Introduction

The assembly of Gondwana is now viewed as a long-standing protracted process that may encompass most of the Neoproterozoic era (e.g., Meert, 2003; Trindade et al., 2006; Vaughan and Pankhurst, 2008). However, reconstructing the paleogeography of this supercontinent in its early stages is a challenging prospect, so far impeded by the scarcity of relevant well-dated paleopoles in almost all cratonic units involved. The Congo craton, a central piece of the Gondwana jigsaw-puzzle, counts only 10 Proterozoic paleopoles between 1300 and 550 Ma (Tohver et al., 2006). These poles mainly belong to two age groups: an older one (800–700 Ma) showing the rapid drift of Congo before the Panafrican orogenic events, and a younger one (550–500 Ma) corresponding to the time of final assembly of Gondwana elements. The main aim of the present work is to provide new data for the poorly known crucial time lag between 700 and 550 Ma. The target is a Neoproterozoic dolerite dyke swarm intrusive in the northwestern part of the Congo craton (Fig. 1), south of the Panafrican belt of central Africa (Bessoles and Trompette, 1980).

2. Geological setting

The northwestern part of the Congo craton is made of an Archaean TTG nucleus, the Ntem complex (Nédélec et al., 1990), covered by Proterozoic series that crop out at the border of Cameroon and the Central African Republic (CAR) (Fig. 1). The Proterozoic series are made of sediments poorly metamorphosed and nearly undeformed. They are intruded by a doleritic dyke swarm west of Nola. A Panafrican nappe has been thrusted toward the south onto the craton. All these Precambrian formations are unconformably covered by the Cretaceous Carnot sandstones (Fig. 1b).

In the Nola region (southwestern part of the CAR), quartzopelitic sediments constitute the so-called Nola series, analogous to the Cameroon Lower Dja series and to the Congo Sembé series (Bessoles

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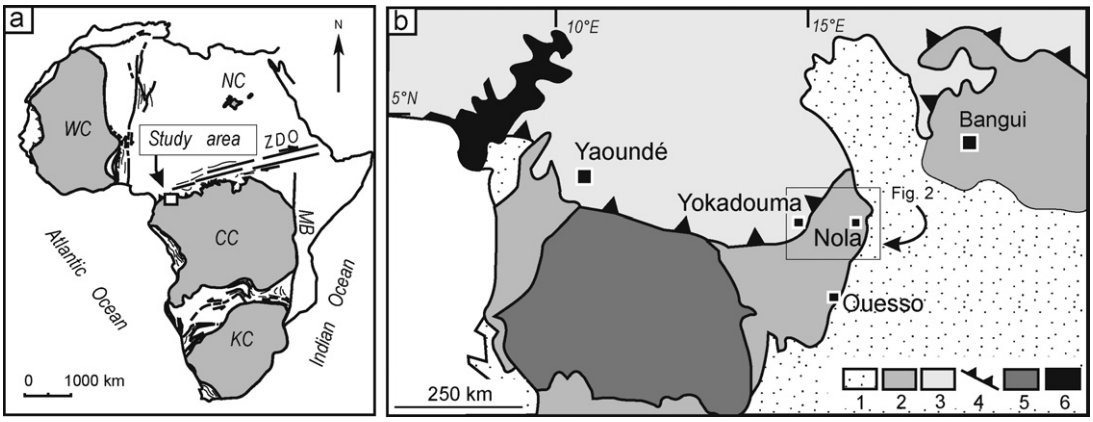


Fig. 1. (a) African cratonic units in grey (WC: West Africa; CC: Congo, KC: Kalahari; NC: Nile), and main tectonic features (CL: Cameroon Line; ZDO: Oubanguides shear zone; MB: Mozambique belt). (b) Geological setting of the Nola region (1) Archean Ntem basement; (2) Proterozoic cover; (3) Panafrican belt; (4) Panafrican thrust; (5) Phanerozoic cover; (6) Recent volcanism. The area of Fig. 2 is delineated by a rectangle.

and Trompette, 1980). The Nola series is made of shales, sandstones and conglomerates. To the north, just south of the Panafrican nappe, the Nola series has been folded and metamorphosed, likely in consequence of the Panafrican thrusting event. This deformed portion was called Bolé-Nord series by Rolin (1992). It is also characterized by interstratified metabasites. In contrast, the southern part of the Nola series is undeformed, except to the east, close to the N–S sinistral Sangha shear zone of probable Late Panafrican age (Fig. 2). Although not directly dated, the sediments of the Nola series are regarded as Paleoproterozoic in age.

Dolerite dikes striking NW–SE form an important mafic intrusive complex crosscutting the Nola series, as well as its equivalent Lower Dja series in Cameroon. They are made of continental tholeiites with low Th/La ratios pointing to an asthenospheric component in their source (Vicat et al., 1996). The Nola dolerites are actually undated, although a pre-Panafrican age is demonstrated by the absence of dikes cutting the nappe. In addition, the metabasites in the Bole-Nord series are likely metamorphosed equivalents of the Nola dolerites. Dolerite pebbles in Cryogenian tillites (Alvarez, 1995) are an additional evidence for an age older than 0.7 Ga for the Nola dike swarm. Vicat et al. (1996) has suggested a correlation with 1 Ga old dolerites found in Gabon, that cannot be ascertained presently.

The Panafrican nappe is called Yaoundé nappe in Cameroon and Oubanguide nappe in CAR. It is a major Himalayan-style tectonic structure, whose age and extension towards the south are still debated. Toteu et al. (1994) proposed an age around 620 ± 10 Ma according to the U–Pb zircon dating of a synkinematically meatmorphosed metabasite near Yaoundé. The nappe contains a diversity of metasediments and some orthoderivate rocks, which reached the beginning of high-pressure granulite facies conditions near Yaoundé (Nzenti et al., 1988). Further south, the base of the nappe was retromorphosed in greenschist facies conditions (Nédélec et al., 1986). East of Yokadouma, the sole mapped thrust corresponds to a pluridecametric level of mylonitic quartzites.

3. New structural data

Foliations and lineations were measured in the study area and its surroundings. They are reported on the map and projection diagrams of Fig. 2. Two different structural patterns are recognized. The formations of Nola series gently dip towards the NW. Mineral lineations, underlined by stretched quartz grains and aligned mica flakes, mainly trend E–W. This is completely different from the broadly N–S trends observed everywhere in the Panafrican nappe (Nzenti et al., 1988; Nédélec et al., 1986).

South of Nola, the Babougué quartzites appear more deformed than the Nola formations, with N–S trending lineations on foliation planes striking N35° to N50° and very gently dipping 10° to 20° to the NW (Fig. 2a and b). Top-to-the-south kinematic indicators were identified in the field. In thin section, these quartzites display large sigma-type quartz grains surrounded by small grains typically resulting from incipient to moderate dynamic recrystallization (Fig. 3a). They are consistent with a top-to-the-south sense of motion, as are some mica fishes and muscovite flakes describing S–C structures (Fig. 3c) characteristic of a non-coaxial deformation. The structural features of the Babougué quartzites are noticeably different from those observed in the Nola region, whereas they are strikingly similar to those of the Panafrican Yokadouma nappe and the Gbaga klippe (Figs. 2 and 3). Since the quartzites of Babougué present a structural pattern similar to that of the Panafrican Yokadouma nappe, we have interpreted them as a klippe of the Panafrican nappe that has been thrusted onto the stable Proterozoic formations of the Congo craton south of Nola (Fig. 2c). The erosion of the nappe left some other klippes, such as the Gbaga klippe to the north of Nola town (Fig. 2c). Our new interpretation implies that the dolerite dyke swarm cropping out east of Nola has been covered by the Panafrican nappe.

In the late-Panafrican Sangha shear zone, quartzites have been also observed. Their foliation plane dips steeply to the west (dips > 55°W) and their mineral lineation plunges gently towards the south.

4. Nola dolerite dikes

The dolerites occur as dikes of metric to decametric thickness, crosscutting the gresopelitic Nola series. Usually, the rocks show well-perserved magmatic textures and have chemical features of typical continental tholeiites (Vicat et al., 1996). A petrographic study enables to distinguish two groups of rocks depending of the alteration intensity: the gently altered dolerites and strongly altered dolerites (Fig. 4).

The gently altered dolerites (NL8 to 10) present mainly subophitic textures. The main mineral phases are phenocrysts of augite (En_{47-35} Fs_{15-29} Wo_{36-41}) and smaller crystals of plagioclase with labrador composition (when unaltered). A brown to green hornblende is locally observed as a late-magmatic mineral. Large automorphic crystals of opaque minerals are always made of ilmenite (Fig. 4a and c), often associated with titanite. Locally,

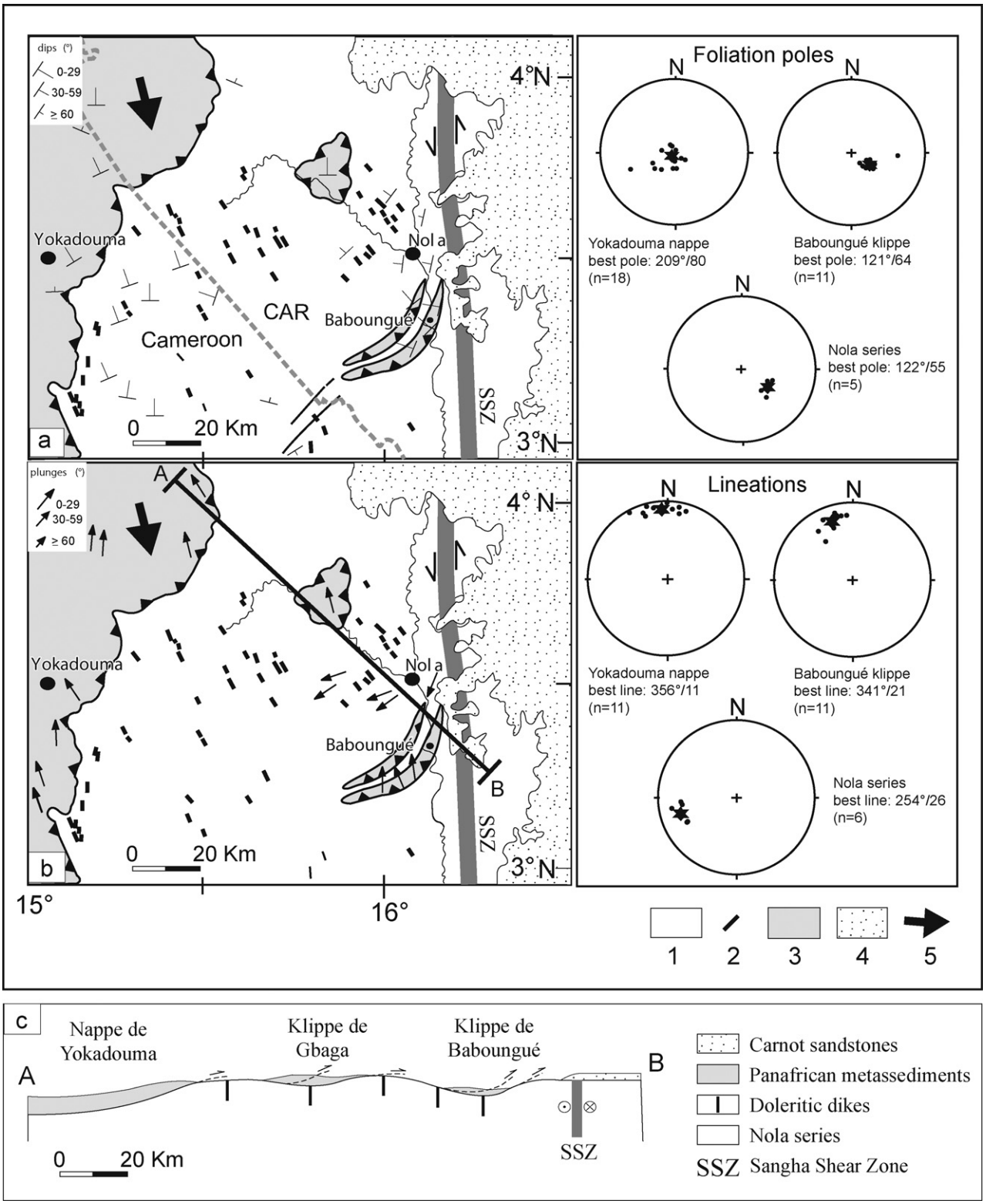


Fig. 2. New structural data for Yokadouma nappe, Baboungué klippe, and Nola region. (a) Foliation map and corresponding stereoplots for each sector. (b) Lineation map and corresponding stereoplots for each sector (SSZ: Sangha Shear Zone (1) Nola Series; (2) Doleritic dikes; (3) Panafrican metassediments; (4) Carnot sandstones; (5) mass transport direction). (c) NW–SE schematic section from the Yokadouma nappe to the Baboungué klippe.

quartz and alkali feldspar form an interstitial granophyric association. The secondary minerals are green to blue-green amphibole and colourless actinote locally developed after altered augite (and brown hornblende), as well as albite and small grains of epidote resulting from the transformation of plagioclase, and also some chlorite in microfractures. The observation of dolerite sample by scanning electron microscope (SEM) coupled with EDS enables to identify microcrysts of magnetite and iron sulfide (pyrrhotite after

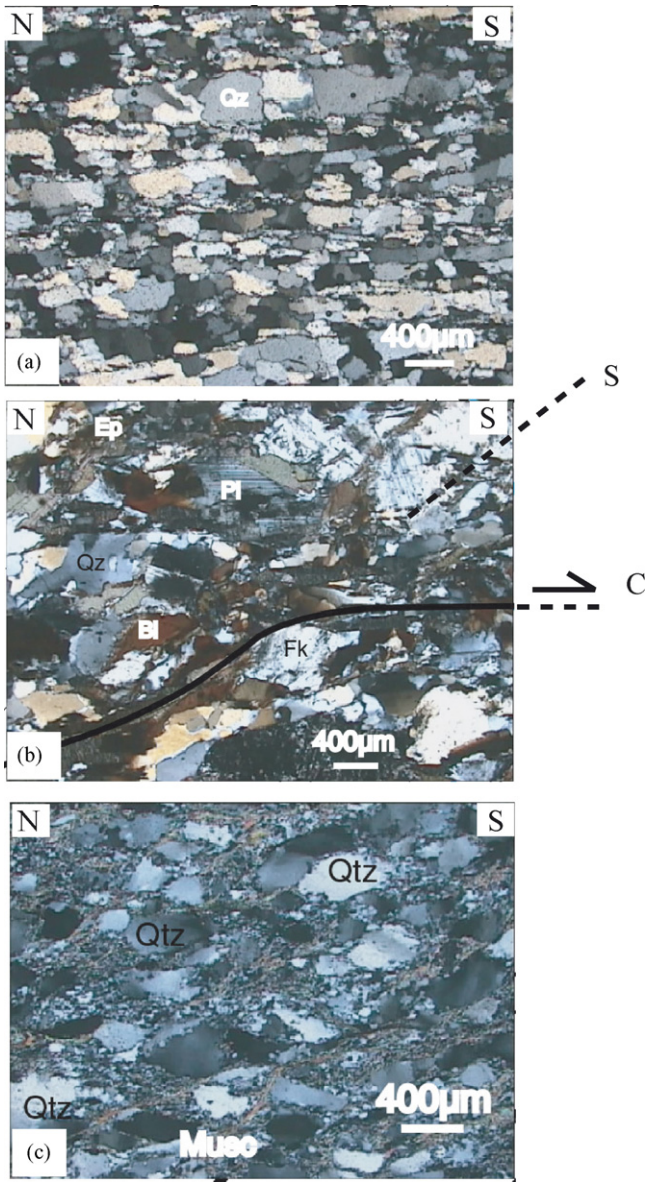


Fig. 3. Microstructures on (a) blastomylonite from the Yokadouma nappe, (b) orthogneiss from the Gbaga klippe, and (c) Baboungué quartzites. Preferential orientation of quartz grains indicate top to the south movement. All sections were cut along the *XZ* plane.

the rock magnetic properties) as tiny inclusions in pyroxene (Fig. 4c and e), as well as an association of pyrrhotite, titanite and chalcopyrite in the secondary amphibole crystals.

The strongly altered dolerites (NL1–NL7) sometimes present still recognizable magmatic textures with relict augite, but most augite phenocrysts are more or less completely uralitized to secondary amphiboles or chlorite with associated small grains of secondary rutile. The rocks are mainly made of secondary minerals typical of greenschist facies conditions: albite, epidote, chlorite and amphibole (bluish green hornblende and actinote). Small grains of rutile are ubiquitous, and especially abundant in the chloritized zones.

These parageneses are typical of hydrothermal imprint in static conditions. The association of actinote, albite, epidote and chlorite points to temperature conditions corresponding to greenschist facies conditions, whereas green to blue-green hornblende occurs in the lower amphibolite facies. Therefore, we consider that the hydrothermal transformations that modified the magmatic minerals occurred at peak-temperatures close to or slightly above the limit between greenschist and amphibolite facies conditions, i.e. around 550 °C.

5. Geochronology

5.1. $^{40}Ar/^{39}Ar$ laser probe dating technique

$^{40}Ar/^{39}Ar$ dating has been applied to single grains of amphibole separated under a binocular microscope after rock crushing, and cleaning in ethanol and distilled water. All grains were packed in aluminium foils and irradiated for 40 h in the McMaster nuclear reactor (Canada) with MMHb hornblende neutron flux monitor dated at 520.4 ± 1.7 Ma (Samson and Alexander, 1987). After irradiation, the single grains were placed on a Cu-holder inside an UHV gas extraction system and baked for 48 h at 200 °C.

Stepwise heating was achieved using a CO_2 laser operating in the continuous mode by regularly increasing its power. The analytical device consists of: (i) a 50 W CO_2 laser; (ii) a beam shutter for selection of exposure times, typically 30 s for individual steps; (iii) divergent and convergent lenses for definition of the beam diameter, typically two times the size of the amphibole grain; (iv) a small inlet line for the extraction and purification of gases; (v) a MAP 215-50 noble gas mass spectrometer. Each analysis involves 5 min for heating and gas cleaning and 8 min for data acquisition by peak switching from mass 40 to 36, through 12 sets of data. System blanks were evaluated every three analyses and yielded values of 2×10^{-12} cm^3 for ^{40}Ar and 3×10^{-14} cm^3 for ^{36}Ar. For each analysis, classical isotope corrections including blanks, mass discrimination, radioactive decay of ^{37}Ar and ^{39}Ar, and irradiation-induced mass interference were applied. The quoted errors represent 1-sigma deviation and were calculated following McDougall and Harrisson (1999). Results are represented as age spectra, Ca/K and $^{36}Ar/^{40}Ar$ vs. $^{39}Ar/^{40}Ar$ isotope correlation plots.

5.2. Amphibole compositions

Microscopic observations already revealed several generations of amphibole. Microprobe analyses were done in Toulouse with the analytic conditions set at 20 A beam current and 15 V accelerating voltage. Analytical uncertainties are less than 1%. Representative analyses are presented in Table 1 and all data are plotted in Fig. 5.

The high temperature brown amphibole is a magnesio-hornblende (rarely an edenite) with XMg = 0.6. Its Ti and F contents are noticeable respectively at about 0.16 and 0.1 apfu. Only the brown hornblende contains F and this is the hallmark of a late-magmatic phase. The large variations of XMgs in the green hornblende, actinote or ferro-actinote are due to variable Fe (III) in the calculated formulae. We consider that only the brown hornblende is of magmatic origin, whereas all other amphiboles (green or blue-green hornblende and actinote) are secondary. Nevertheless, none of these brown magmatic hornblende crystals escaped partial modification, because they always display some colour zoning from a brown core to a green or bluish green rim. For sake of Ar–Ar dating, we selected two samples with recognizable brown magmatic hornblende: NL1 and NL8.

5.3. Ages

As shown in Fig. 6, the two samples NL1 and NL8 display very distinct isotopic and age characteristics. NL1 has a saddle-shaped age spectrum with maximum ages close to 800 Ma and a minimum age of 595 ± 7 Ma. The K/Ca diagram is strongly discordant with a

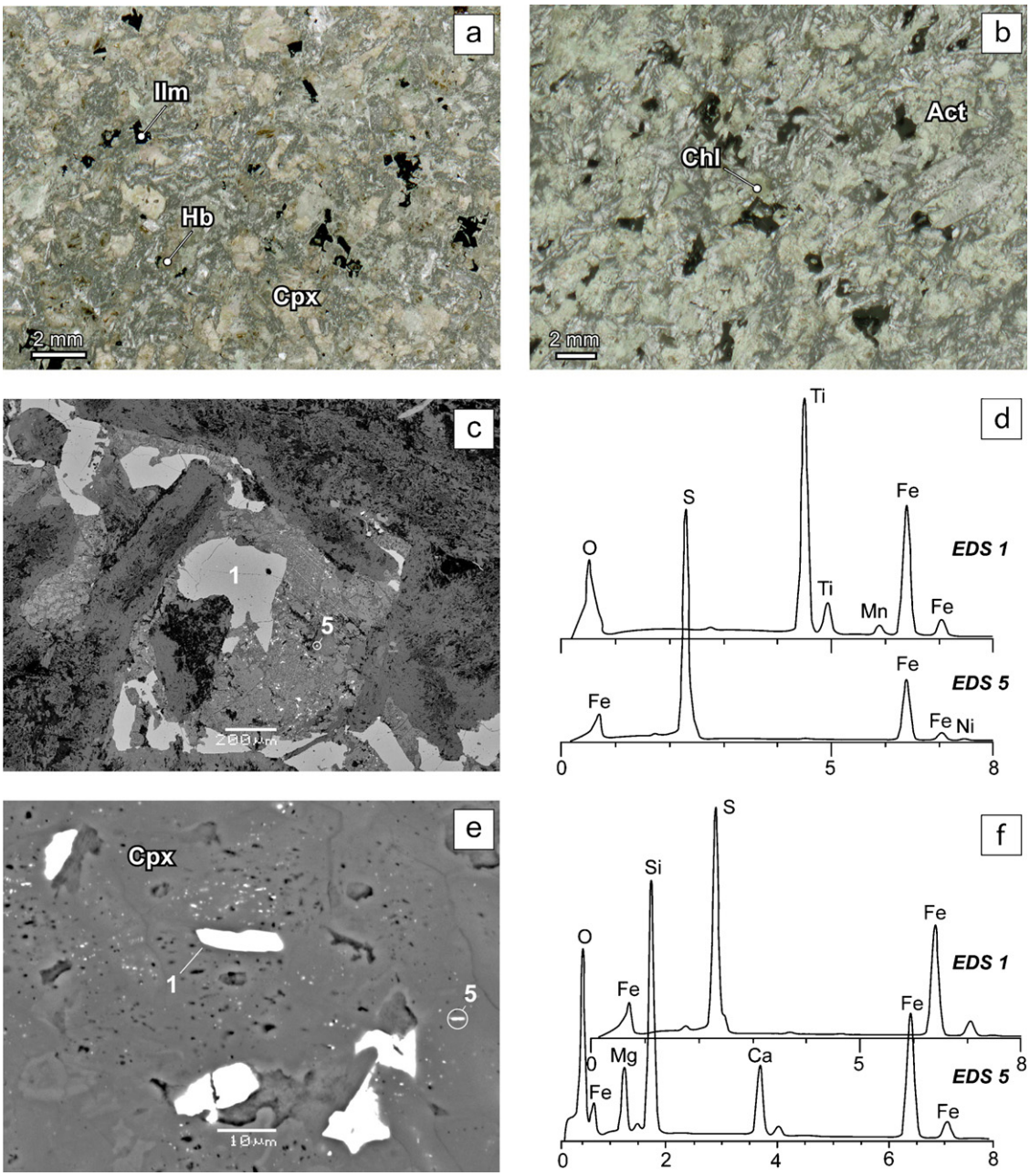


Fig. 4. Petrography of Nola dikes. Above: photomicrographs of (a) slightly altered and (b) strongly altered dolerite. Below: SEM images of (c) major mineral components with large ilmenite grain (1) in the center and small sulphide inclusions (5), and (e) detail of pyroxene grain with inclusions of pyrrhotite (1) and magnetite (5). The right panel (d and f) shows the corresponding EDS spectra.

general decrease of ratios from low to high argon extraction temperatures and no correlation with the age variations. In the $^{36}Ar/^{40}Ar$ against $^{39}Ar/^{40}Ar$ isotope correlation plot, the data are scattered along a line with a large MSWD value and meaningless intercepts. By contrast, sample NL8 has a less discordant age spectrum with a plateau date of 570.3 ± 5.8 Ma related to 80% of the gas released and a younger age of 539.7 ± 5.7 Ma for the first heating step. The K/Ca ratios display a decreasing trend similar to that recorded by sample NL1, which may be due to chemical heterogeneities in both amphiboles, in agreement with the observed colour zonation. Sample NL8 has an intercept age of 570.8 ± 6.5 Ma in the isotope correlation plot, with an atmospheric composition of initial argon (MSWD = 1.25). Table 2 also indicates that the Cl/K ratios (imaged by $^{38}Ar/^{39}Ar$) are significantly lower in sample NL8 than in sample NL1.

According to similar age spectra reported in the literature (e.g. Harrison and McDougall, 1981), the saddle-shaped pattern of sample NL1 can be interpreted to reflect the presence of excess argon trapped in the lattice of amphibole. The relatively high Cl/K ratios reported for this amphibole suggest that excess argon was probably introduced by chlorine rich fluids that circulated at the time the argon chronometer became a closed system. The minimum age of 595 ± 7 Ma recorded by the intermediate portion of the spectrum is interpreted as a maximum age for the closure of amphibole to argon diffusion. The best estimate of amphibole closure age in the Nola dolerites is provided by the plateau and intercept ages of sample NL8 at 571 ± 6 Ma. This amphibole does not contain any evidence of excess argon contamination and its age of 571 Ma is interpreted to date dolerite cooling at about 550 °C (Dahl, 1996).

Table 1
Microprobe data for amphiboles from Nola dolerites

Analysis no.	Brown hornblende			Green to blue-green hornblende			Actinote	
	#1	#2	#6	#3	#4	#5	#7	#8
SiO_2 (%)	46.3	46.59	46.38	47.76	46.07	50.05	54.26	53.41
TiO_2	1.53	1.47	1.5	0.47	0.04	0.13	0.06	0
Al_2O_3	6.06	5.46	5.71	3.92	4.62	2.65	1.63	1.60
Cr_2O_3	0	0	0.04	0	0.01	0	0.19	0.15
Fe_2O_3(c)	5.33	5.56	5.65	8.49	3.11	4.64	0.42	0.51
FeO(c)	13.4	12.96	12.53	18.78	28.83	22.14	12.85	14.01
MnO	0.31	0.24	0.18	0.32	0.31	0.66	0.34	0.35
MgO	11.61	11.97	11.95	6.76	2.39	6.07	14.86	13.98
CaO	10.75	10.69	10.56	9.96	10.76	10.4	12.75	12.6
Na_2O	2.23	2.14	2.06	1.06	1.26	0.71	0	0.09
K_2O	0.87	0.84	0.92	0.33	0.31	0.13	0.06	0.05
NiO	0.07	0	0.03	0	0.01	0.13	0.11	0.04
F	1.49	2.23	2.17	0	0	0	0	0
Cl	0.33	0.31	0.28	0.29	0.23	0.1	0.04	0
H_2O(c)	1.22	0.87	0.9	1.89	1.83	1.93	2.07	2.05
O=F	0.63	0.94	0.92	0	0	0	0	0
O=Cl	0.07	0.07	0.06	0.06	0.05	0.02	0.01	0
Sum oxides	100.78	100.32	99.89	99.97	99.72	99.74	99.62	98.85
Si	6.884	6.947	6.931	7.289	7.319	7.655	7.841	7.828
Ti	0.171	0.164	0.169	0.054	0.005	0.015	0.007	0
Al IV	1.062	0.959	1.006	0.706	0.681	0.345	0.159	0.172
Al VI	0	0	0	0	0.184	0.133	0.118	0.105
Cr	0	0	0.005	0	0.001	0	0.021	0.018
Fe^{3+}	0.597	0.623	0.635	0.975	0.372	0.535	0.046	0.056
Fe^{2+}	1.666	1.616	1.566	2.398	3.831	2.832	1.553	1.718
Mn^{2+}	0.039	0.031	0.022	0.042	0.041	0.085	0.042	0.043
Mg	2.573	2.660	2.662	1.537	0.565	1.383	3.200	3.055
Ca (B)	1.712	1.708	1.690	1.628	1.832	1.704	1.975	1.979
Na (B)	0.288	0.292	0.310	0.372	0.168	0.296	0.025	0.021
Na (A)	0.353	0.327	0.288	0	0.221	0	0	0.004
K (A)	0.165	0.160	0.175	0.064	0.062	0.026	0.011	0.010
Ni	0.008	0	0.004	0	0.001	0.016	0.013	0.005
F	0.702	1.054	1.028	0	0	0	0	0
Cl	0.083	0.078	0.071	0.074	0.063	0.027	0.009	0
OH	1.215	0.869	0.902	1.926	1.937	1.973	1.991	2
Sum of cations	17.518	17.487	17.463	17.005	17.283	16.942	16.986	17.014
XMg	0.61	0.62	0.63	0.39	0.13	0.33	0.67	0.64

Table 2
$^{40}Ar/^{39}Ar$ analytical results

Step	40/39	38/39	37/39	36/39 (E-3)	$F^{39}Ar$ released	%40*	40*/39K	Age (Ma)	±1 S.D. (Ma)
NL1 amphibole	$J = 0.009954$								
1	910.292	1.164	2.8118	2824.568	0.81	8.33	75.97	1015.9	568.5
2	684.969	0.638	2.3898	2113.448	5.18	8.85	60.70	852.7	162.1
3	122.078	0.319	2.4572	249.218	17.18	39.81	48.68	712.8	18.6
4	52.140	0.246	3.2523	24.947	28.15	86.30	45.09	668.9	5.8
5	43.558	0.274	8.3756	11.670	39.30	93.49	40.95	616.8	4.6
6	42.439	0.269	7.0385	13.458	45.91	91.84	39.15	593.7	9.2
7	42.415	0.280	6.7380	12.940	58.96	92.14	39.25	595.0	6.6
8	42.251	0.253	5.4851	6.813	67.01	96.17	40.78	614.6	6.1
9	43.405	0.320	8.2823	0.012	70.21	99.96	44.25	658.4	8.1
10	61.526	0.376	52.3368	37.786	100.00	88.21	56.19	801.4	5.6
Total age									703.0 ± 11.5
NL8 amphibole	$J = 0.009954$								
1	48.674	0.087	0.9043	46.412	18.14	71.93	35.03	539.7	5.7
2	38.161	0.074	4.5035	3.904	51.99	97.82	37.44	571.5	2.4
3	38.399	0.082	2.5673	2.255	67.00	98.73	37.97	578.4	4.5
4	38.204	0.069	2.5008	3.047	73.32	98.09	37.54	572.7	3.5
5	37.739	0.072	3.0409	4.523	80.13	97.02	36.69	561.6	4.4
6	41.842	0.139	10.7049	17.297	83.93	89.67	37.78	575.9	8.3
7	39.591	0.105	12.6531	12.955	100.00	92.69	37.00	565.7	2.6
Total age									565.7 ± 5.1

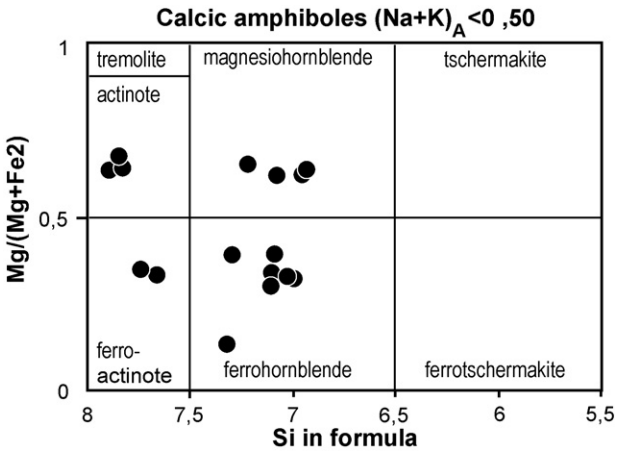


Fig. 5. Composition of amphiboles from Nola dikes.

6. Paleomagnetism

6.1. Sampling and methods

Paleomagnetic sampling was performed in few sites due the poor outcropping conditions in the densely forested Nola region. Most dikes crop out as a series of aligned boulders, for which the original orientation is not certain. However, in a few outcrops we have found dikes with well-preserved margins consistently oriented in the NW–SE direction, indicating that these rocks were *in situ*. Drill-cored paleomagnetic samples were collected at 10 of those dikes (10 sites) in two sectors of the Nola region (Fig. 7). They were oriented in the field by sun and magnetic compass.

In the laboratory, samples were cut into 22 mm × 25 mm cylindrical specimens, and submitted to stepwise thermal and alternating field (AF) treatments. Rock chips from the same samples were used for thermomagnetic curves and optical and electronic (SEM) microscopy. Thermomagnetic curves were performed in a KLY3-CS3 Agico device. AF demagnetization was performed on 76 specimens along 16–29 steps. Thermal demagnetization was performed on 23 specimens along 16 steps. Magnetic remanence was measured using JR5/6A Agico spinner magnetometers or a 2G cryogenic magnetometer. Alternating field (AF) demagnetization was

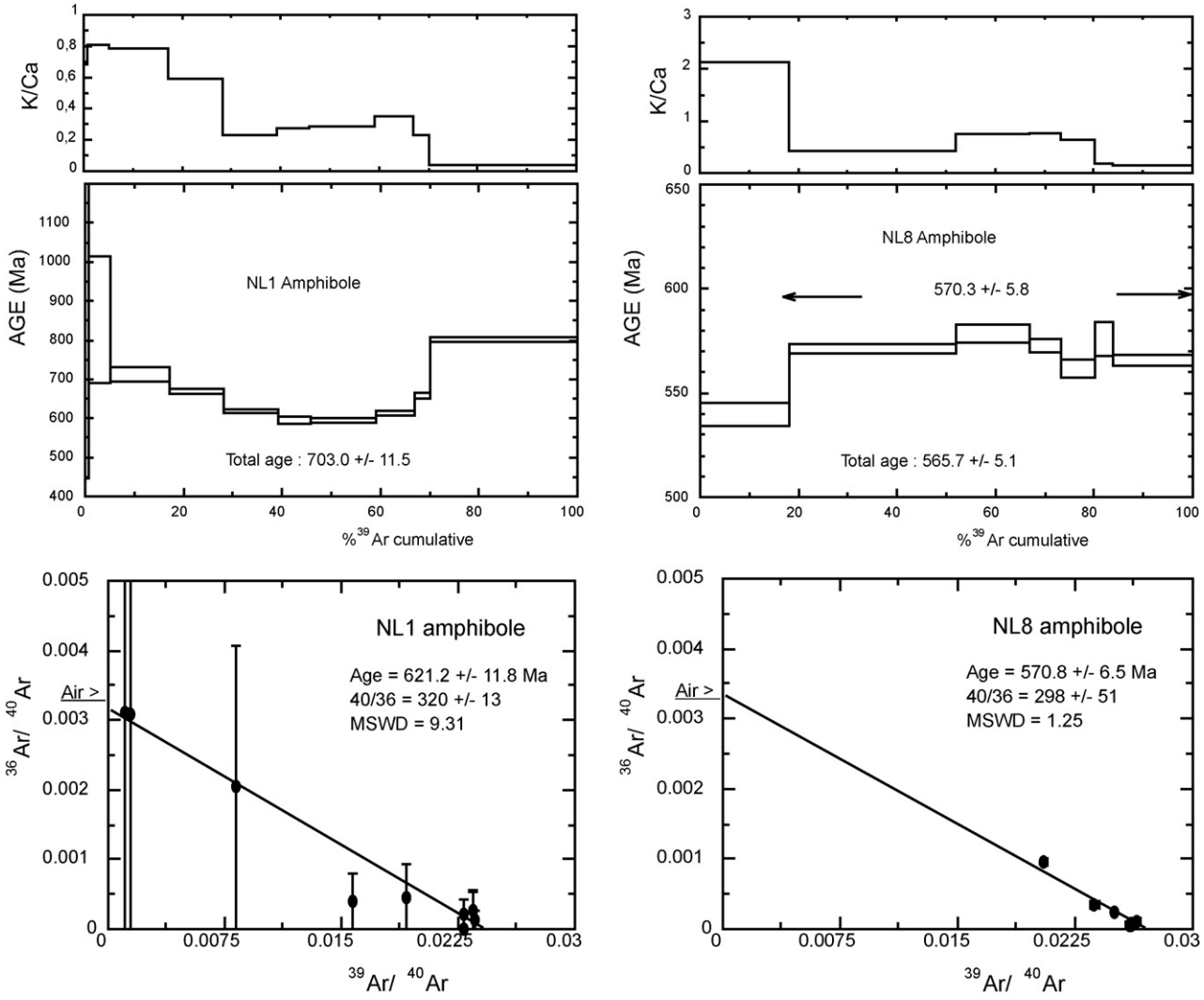


Fig. 6. Age spectra, Ca/K and isotope correlation plots for Nola amphiboles.

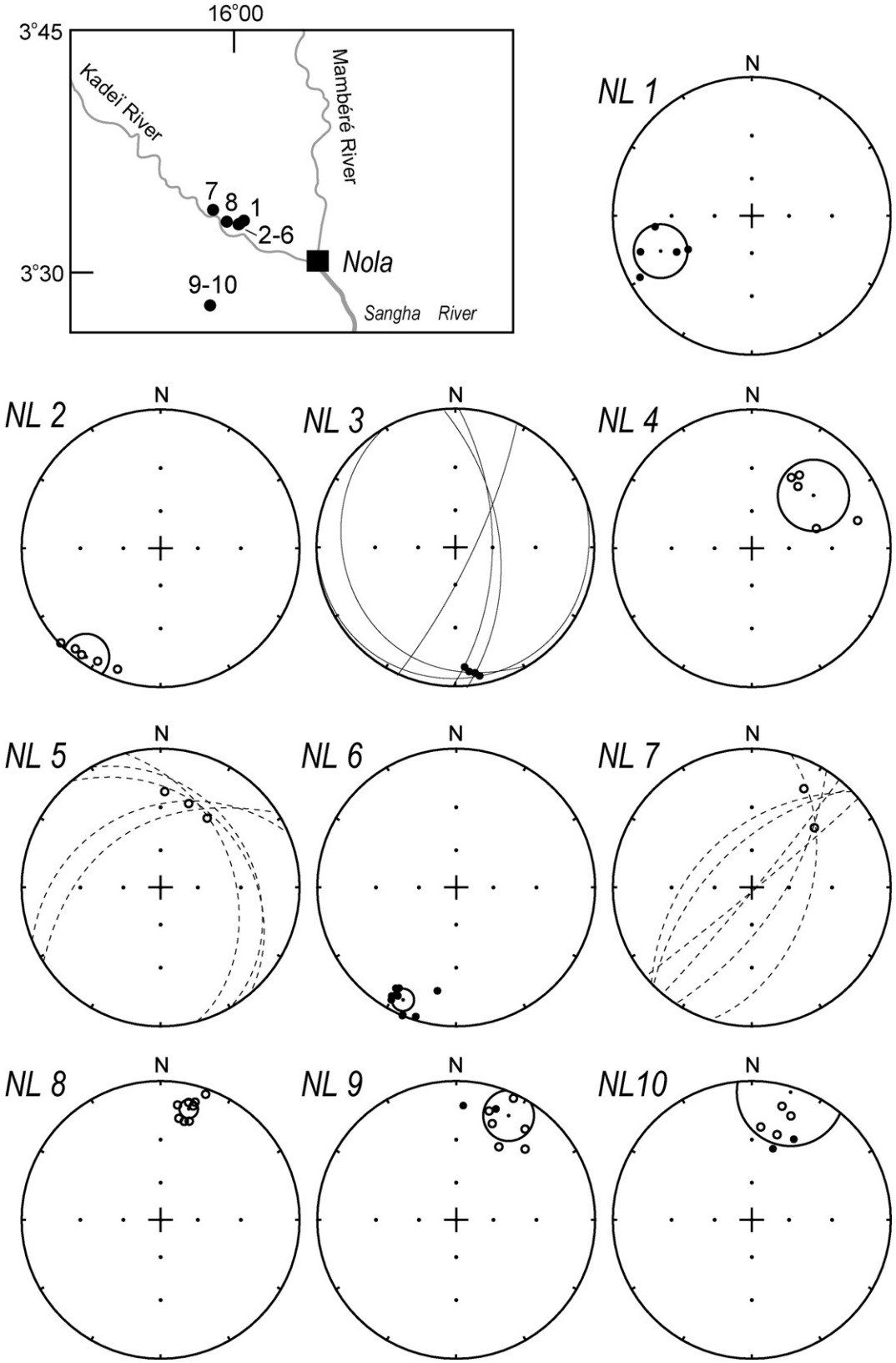


Fig. 7. Location of paleomagnetic sites, and stereoplots with characteristic directions isolated for each sampling site. Means for sites NL3, NL5 and NL7 were obtained through great circles intersection. Negative (positive) vectors are represented by empty (full) circles.

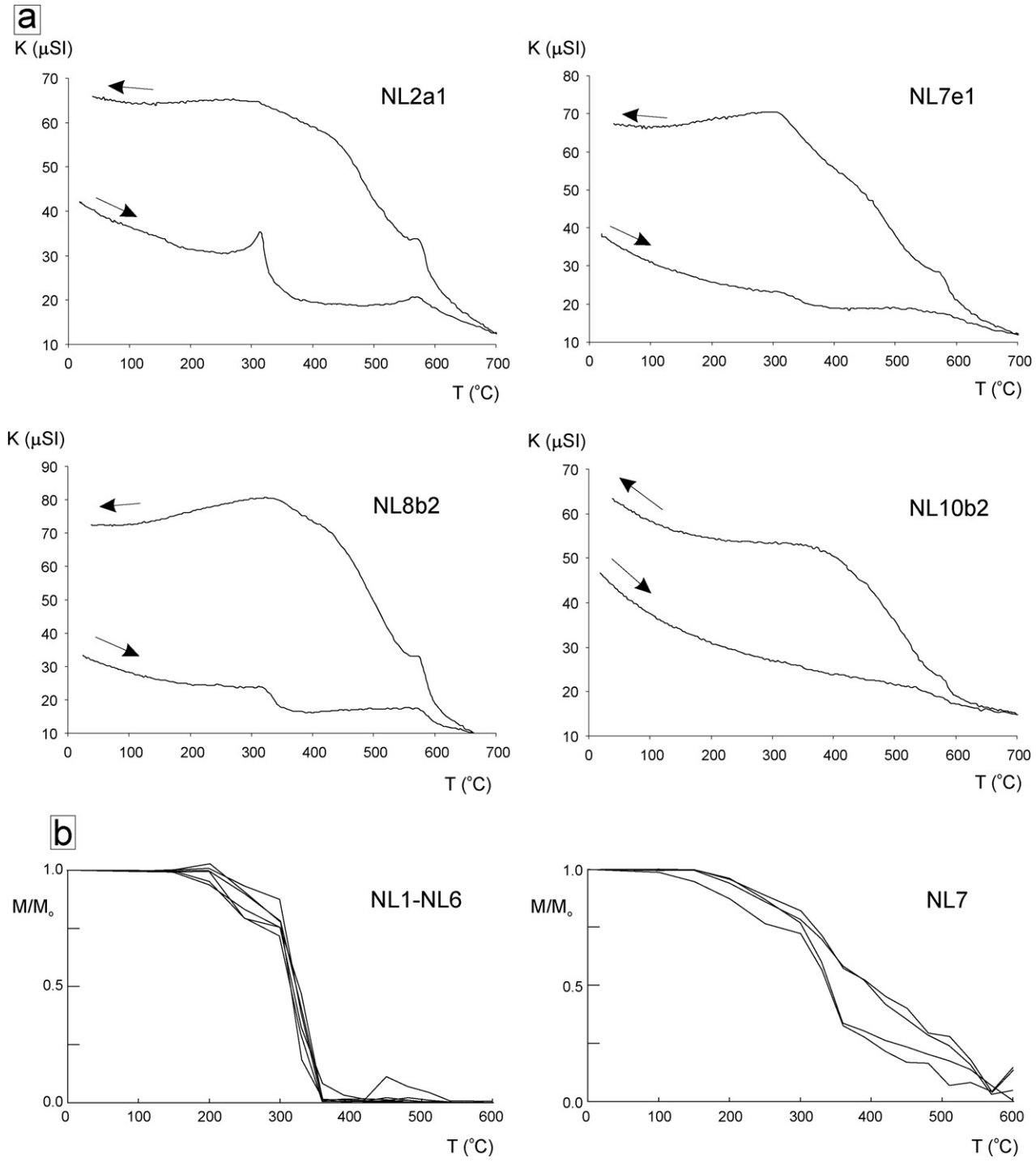


Fig. 8. (a) Thermomagnetic curves for Nola samples, showing the mixture of pyrrhotite and magnetite (NL2a1, NL7e1, NL8b2) and only magnetite (NL10b2). Magnetization decay curves during thermal treatment for sites NL1–NL6 with pyrrhotite as the main carrier and site NL7 with pyrrhotite and magnetite as the main carriers.

performed with an AMU1 Agico tumbler-demagnetizer or by an automated three-axis AF-demagnetizer coupled with the cryogenic magnetometer. Thermal demagnetization was performed with a MMTD60 oven (peak temperatures within ±2 °C) with heating time of 40 min for each step. All these equipments are housed into magnetically shielded rooms. Magnetic components for each specimen were identified by end-points and great circles in orthogonal plots, and calculated using principal component analysis (Kirschvink, 1980). Fisherian statistics was used to calculate vector mean directions and paleomagnetic poles.

6.2. Magnetic mineralogy

Thermomagnetic curves were performed for one sample per site. The curves are not reversible, due to significant mineralogical transformation during heating (Fig. 8a–d). For most sites heating curves reveal two transitions, at 320 and 580 °C, indicating the presence of pyrrhotite and magnetite as the main magnetic minerals in Nola dolerites (samples NL2a1, NL7e1 and NL8b2). In sample NL10b2 only the 580 °C magnetite transition is observed. These results are corroborated by thermal demagnetization decay curves

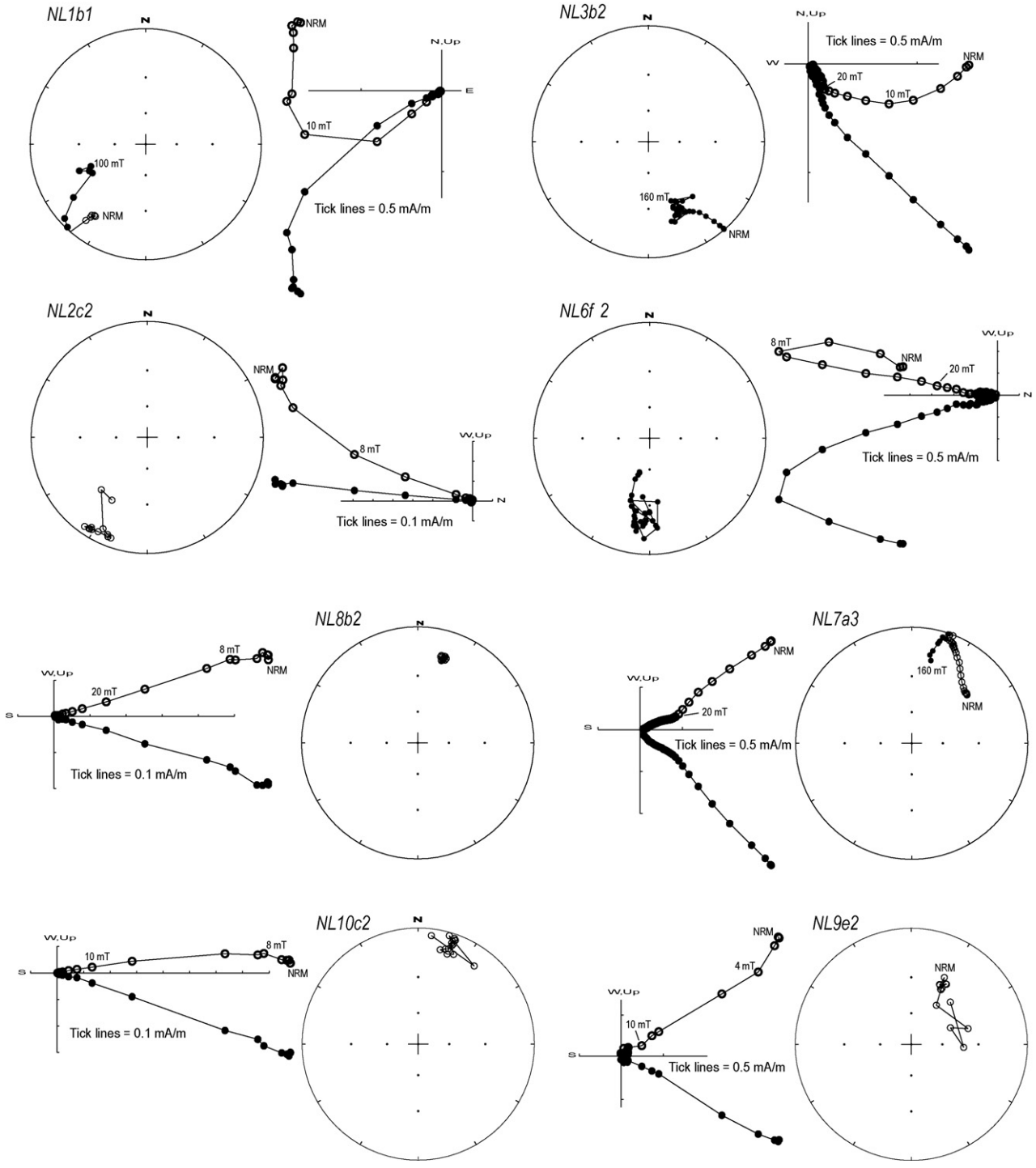


Fig. 9. Demagnetization behavior of Nola samples from stereoplots and orthogonal diagrams. Empty (full) circles in stereoplots represent upward (downward) directions, whereas empty (full) symbols in orthogonal plots indicate vertical (horizontal) projections.

(Fig. 8e and f). In sites NL2–NL6 pyrrhotite is the main magnetic carrier whereas in site NL7 both magnetite and pyrrhotite carry the characteristic magnetic component.

The presence of pyrrhotite and magnetite, inferred from thermomagnetic curves and thermal demagnetization patterns, is confirmed by SEM observations (Fig. 4c and e). Pyrrhotite occurs as small (~10 μm) crystals associated to the ubiquitous, large ilmenite crystals that correspond to the major opaque phase in Nola dolerites (Fig. 4c). Low-Ti magnetite occurs as very fine-grained, elongated inclusions in pyroxene crystals (Fig. 4e).

6.3. Magnetic components and paleomagnetic pole

Two nearly antipodal components were isolated in nine sites by thermal and AF demagnetizations after removal of a weak, randomly oriented viscous component at 8–10 mT and temperatures

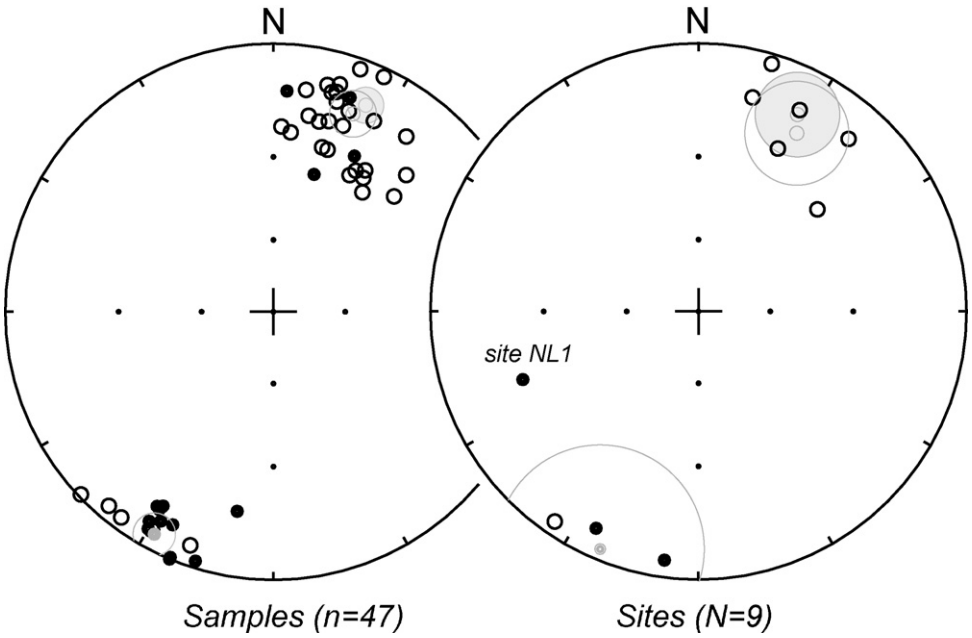


Fig. 10. Mean directions for (a) samples and (b) sites. Empty grey circles correspond to the means for each component, whereas full grey circles indicate the mean after inversion of the southwestern component.

above 150 °C (Fig. 9). A shallow south to southwest component was obtained on sites NL2, NL3 and NL6, whereas a shallow northeast component was obtained on sites NL4, NL5, NL7, NL8, NL9 and NL10 (Figs. 7 and 9). In sites NL3, NL5 and NL7 only few end-points were obtained. Given the curved nature of their demagnetization patterns we have estimated their mean direction through great circles using the method of McFadden and McElhinny (1988). Results for these sites are coherent with those obtained by end-points in the other sites (Fig. 7). Site NL1 has shown a different west to southwest directed component and was not retained for the mean calculation (Fig. 7).

After inversion of site mean directions with the south to southwest component, they lie close to the northeast component isolated in the other sites (Fig. 10). They match the criteria for a positive site-based reversal test (McFadden and McElhinny, 1990), suggesting they represent normal and reverse polarities of the same averaged magnetic field. Mean direction and VGP calculated for each site, as well as the mean direction and corresponding paleomagnetic pole for Nola are given in Table 3. Mean directions were obtained for sites and samples (Fig. 10). The pole obtained after combination of sites with southwest and northeast directions lies at 304.8°E and 61.8°S ($dp = 5.4$, $dm = 10.7$). This pole is similar to that obtained after the sample-mean average direction that lies at 305.1°E and 64.5°S ($dp = 2.3$, $dm = 4.5$).

7. Discussion

7.1. Age of the Nola paleomagnetic pole

The minerals that carry the magnetization in the dolerite dikes of Nola are pyrrhotite and magnetite which are secondary minerals crystallized during hydrothermal alteration of the iron-bearing silicates. Some of these minerals, such as the tiny magnetite inclusions in the pyroxenes, may have formed very early during cooling at the end of the magmatic stage. However, sulfide development is more typical of a hydrothermal greenschist imprint occurred after the crystallization of the dikes. Both magnetite and pyrrhotite yield the same magnetization in spite of their unblocking temperatures

Table 3
Paleomagnetic results

Site	*n*/*N*	Dec	Inc	α95	*R*	*K*	VGP lat	VGP lon	Dp	Dm
NL1	5/6	248.8	19.8	14.7	4.86	28.20	−20.1	298.2	8.0	15.4
NL2	5/6	214.4	−3.0	10.0	4.93	59.84	−55.6	288.5	5.0	10.0
NL3	6/8	187.8	3.8	8.1	Great circles		−80.5	320.6	4.1	8.1
NL4	5/6	49.4	−29.4	21.4	4.71	13.72	−37.5	308.9	13.1	23.6
NL5	5/6	26.0	−21.9	22.4	Great circles		−60.2	316.2	12.5	23.7
NL6	10/10	205.3	6.4	5.1	9.90	91.10	−63.8	300.6	2.6	5.1
NL7	7/10	41.0	9.1	12.8	Great circles		−48.2	296.9	6.5	12.9
NL8	8/8	14.2	−11.1	4.6	7.95	148.67	−73.1	318.6	2.4	4.7
NL9	9/10	26.7	−9.9	12.1	8.58	18.97	−62.0	303.5	6.2	12.2
NL10	6/6	16.5	−2.2	22.0	5.51	10.22	−72.5	301.1	11.0	22.0
Mean C1	3	202.5	2.4	22.1	2.94	32.04	−67.0	297.4	11.1	22.1
Mean C2	6	28.5	−14.2	13.8	5.80	24.47	−59.6	306.7	7.2	14.1
C1 + C2 (sites)	9/10	26.7	−11.2	10.5	8.67	24.01	−61.8	304.8	5.4	10.7
C1 + C2 (samples)	47/76	24.1	−9.7	4.5	44.94	2.35	−64.5	305.1	2.3	4.5

Note: The mean ChRM is given by its declination (Dec), inclination (Inc), radius of the 95% confidence cone (α95), and the parameter *R* and *K* from Fisherian statistics. *n*/*N* indicates the number of samples considered for the mean (*n*) and the number of samples measured. The latitude and longitude of the virtual geomagnetic poles are given by VGP lat and VGP lon. Dp and Dm are the confidence parameters for the VGP.

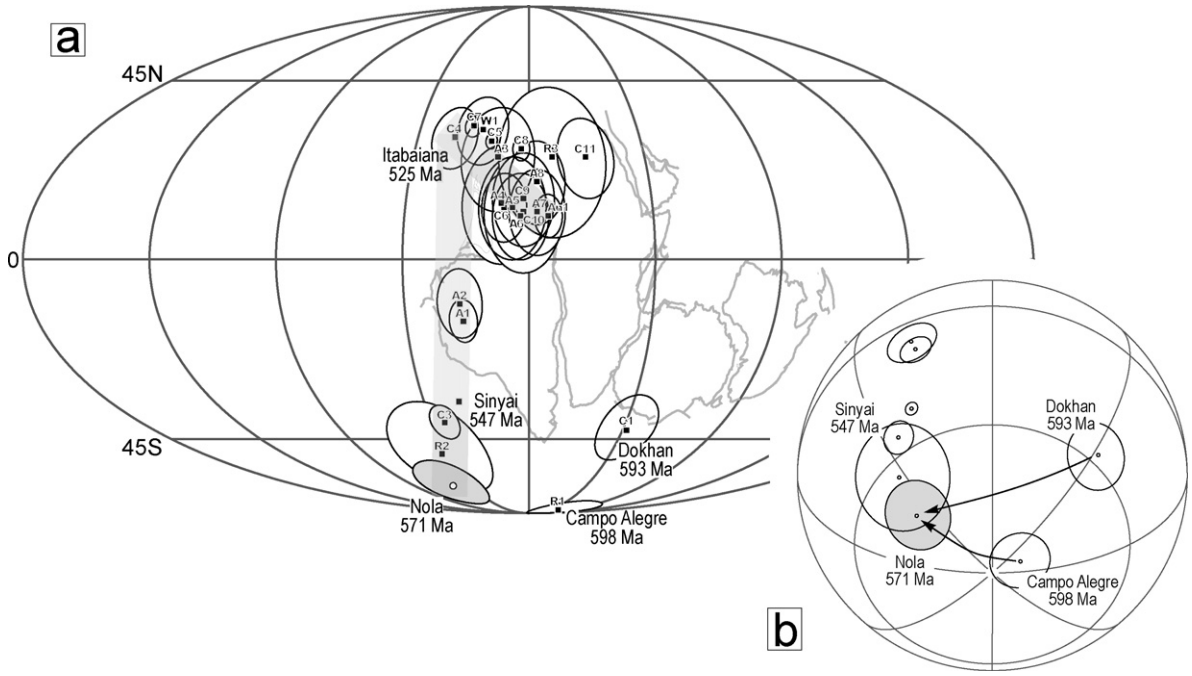


Fig. 11. Apparent polar wander path for the West Gondwana. (a) Nola paleopole and reference poles numbered after Trindade et al. (2006) using the "tight-fit" reconstruction parameters that take into account pre-Mesozoic deformation within the African plate. (b) Inset with orthogonal projection showing the southern hemisphere with Nola, Dokhan, Sinyai and Campo Alegre poles and possible APW paths following these paleopoles.

being significantly different, thus strengthening our assumption that the magnetization was likely acquired during a hydrothermal imprint. The secondary mineral parageneses were acquired in the range of the Curie temperature for magnetite or slightly below, thus characterizing a thermoremanence or a thermochemical remanence. Any primary magnetization eventually recorded in these dikes was probably erased by the hydrothermal event.

The hydrothermal event also occurred at a temperature slightly above the Ar–Ar closure. Hence, the age of the hydrothermal event is now well-establish at about 571 Ma, which is also the age of the Nola magnetic pole. As the dolerite dykes were emplaced at a minimum age of 700 Ma, this hydrothermal event is not the consequence of their emplacement at high-level in the unmetamorphosed Nola series. Rather, this age is evocative of a Panafrican event. Since our new structural data demonstrate that the Panafrican nappe was originally thrusted south of the Nola, we attribute the hydrothermal imprint to fluid circulation at the base of the nappe. Incidently, our new Ar–Ar age may provide a better age constrain for the thrusting, suggesting it is younger than the 620 Ma U–Pb age obtained so far from a deformed metabasite by Toteu et al. (1994). The 620 Ma age may be reinterpreted as a crystallization age for this magmatic rock and not the age of the tectonic event. Alternatively, the 571 Ma age may represent metamorphic retrogression at the base of the nappe. We know that metamorphism reaches amphibolite to granulite conditions further north in the nappe, and the base of the nappe has then retrogressed to greenschist facies conditions in Cameroon (Nédélec et al., 1986).

7.2. Neoproterozoic apparent polar wander path (APWP) for the Congo craton

The Nola pole provides a good late Neoproterozoic constraint to the apparent polar wander path (APWP) of the Congo craton. This pole satisfies six ($Q = 6$) out of the seven criteria of Van Der Voo (1990): (1) the age of magnetization was determined from Ar to Ar dating of amphiboles in one of the dykes (NL8), being interpreted as the age of the metamorphic imprint on these rocks responsible for their thermal or thermochemical remanence; (2) despite the limited number of sites (9) and samples (47) the occurrence of reversions and the good agreement of site-mean and sample-mean poles both with moderate dispersion, suggest the secular variation has been averaged out; (3) in all considered sites, coherent remanence vectors were isolated by principal component analysis after detailed stepwise AF and thermal demagnetization in all studied sites; (5) the last event affecting Nola dykes is the metamorphic imprint of the Panafrican nappe at 570 Ma, when its magnetization was reset; (6) the Nola pole does not coincide with younger poles for the Congo craton or for the African plate; (7) southwest and northeast mean directions pass a reversal test, indicating they represent normal and reverse polarities of the same averaged magnetic field. The lack of field stability tests, such as the baked contact test or the fold test is the only criteria not fulfilled in the Q-index classification.

The 571 Ma Nola pole is the only well-dated pole available for Gondwana in the 580–550 Ma interval (Fig. 11). It bridges the gap between the Dokhan volcanics pole dated at 593 ± 15 Ma (Davies et al., 1980; Wilde and Youssef, 2000) and the Sinyai dolerite pole dated at 547 ± 4 Ma (Meert and Van Der Voo, 1996). The Dokhan volcanics erupted during the transition from collision-related magmatism to extension-related magmatism in northeastern Egypt, marking the end of accretionary processes at the Arabian Nubian Shield in the eastern border of the Congo craton (e.g., Eliwa et al., 2006). Therefore, the Dokhan pole can be used to constrain the paleogeographic position of the Congo craton. The Sinyai dolerite pole was recorded during a remagnetization event coeval to Panafrican greenschist metamorphism that affected metassediments and the intruding dolerites in central Kenya (Meert and Van Der Voo, 1996). Alternatively, an apparent polar wander track could be traced between the pole obtained on the Campo Alegre volcanics (598 ± 29 Ma; D'Agrella-Filho and Pacca, 1988; Basei et al., 1998), the Nola dikes pole, and the Sinyai dolerite. In both cases, the Nola pole marks a strong change in direction of the Congo APWP, which

assumes an elbow shape between 590 and 550 Ma (Fig. 11). If the apparent polar movement is attributed only to the translation of the Congo craton, the APWP suggested here would imply moderate plate velocity between 593 and 571 Ma (Dokhan-Nola: average speed of 22 cm/year) and between 571 and 547 Ma (Nola-Sinyai: average speed of 12 cm/year), with a significant increase in plate velocity after 550 Ma reaching up to 35 cm/year.

8. Summary and conclusions

We report new structural, geochronological and paleomagnetic data for dolerite dikes from the Nola region (Central African Republic), which is situated at the northern border of the Congo craton. Our results indicate that the Panafrican nappe has covered the Nola region, promoting pervasive hydrothermal greenschist metamorphism in the underlying cratonic basement. This metamorphic event has also promoted pervasive remagnetization of the Nola dolerite dikes. Both metamorphism and magnetic resetting were dated by the $^{40}Ar/^{39}Ar$ method on amphibole grains from the dikes at 571 ± 6 Ma. The late Neoproterozoic dual-polarity magnetization obtained on these dikes corresponds to a paleopole at 304.8°E and 61.8°S ($dp = 5.4$, $dm = 10.7$) graded at $Q = 5$. This pole, together with other key-poles from the Congo craton, suggests an elbow-shaped APWP between 593 and 547 Ma at the waning stages of the Panafrican orogenic events in northern Congo craton.

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References

Alvarez, P., 1995. Evidence for a Neoproterozoic carbonate ramp on the northern edge of the Central African craton: relations with late proterozoic intracratonic troughs. Geologische Rundschau 84, 636–648.

Basei, M.A.S., Citroni, S.B., Siga Jr., O., 1998. Stratigraphy and age of Fini-Proterozoic Basins of Paranà and Santa Catarina States, Southern Brazil. Bol. Inst. Geociènc. 29, 195–216.

Bessoles, B., Trompette, R., 1980. Geologie de l'Afrique: la chaine panafricaine-zone mobile d'Afrique centrale (partie sud) et zone mobile soudanaise. Memoire du Bureau de Recherches Geologique et Minieres 92, 397.

D'Agrella-Filho, M.S., Pacca, I.G., 1988. Palaeomagnetism of the Itajai, Castro and Bom Jardim groups from southern Brazil. Geophys. J. 93 (2), 365–376.

Dahl, P.S., 1996. The effects of composition on retentivity of argon and oxygen in hornblende and related amphiboles: a field-tested empirical model. Geochim. Cosmochim. Acta 60 (19), 3687–3700.

Davies, J., Nairn, A.M., Ressetar, R., 1980. The paleomagnetism of certain Late Precambrian and Early Paleozoic rocks from the Red Sea Hills, Eastern Desert, Egypt. J. Geophys. Res. 85 (B7), 3699–3710.

Eliwa, H.A., Kimura, J.-I., Itaya, T., 2006. Late Neoproterozoic Dokhan Volcanics, North Eastern Desert, Egypt: geochemistry and petrogenesis. Precambrian Res. 151 (1–2), 31–52.

Harrison, T.M., McDougall, I., 1981. Excess Ar in metamorphic rocks from Broken Hill, New South Wales: implications for Ar/Ar age spectra and the thermal history of the region. Earth Planetary Sci. Lett. 55 (1), 123–149.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astronomical Soc. 62 (3), 699–718.

McDougall, I., Harrisson, T.M., 1999. Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method. Oxford University Press, Oxford, p. 269.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism. Earth Planetary Sci. Lett. 87 (1–2), 161–172.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103 (3), 725–729.

Meert, J.G., 2003. A synopsis of events related to the assembly of the eastern Gondwana. Tectonophysics 362 (1–4), 1–40.

Meert, J.G., Van Der Voo, R., 1996. Paleomagnetic and Ar/Ar study of the Sinyai Dolerite, Kenya: implications for Gondwana assembly. J. Geol. 104 (2), 131–142.

Nédélec, A., Macaudiere, J., Nzenti, J.P., Barbey, P., 1986. Evolution structurale et metamorphique des schistes de Mbalmayo (Cameroun). Implications pour la structure de la zone mobile pan-africaine d'Afrique central, au contact du craton du Congo. Comptes Rendus—Academie des Sciences, Ser. II 303 (1), 75–80.

Nédélec, A., Nsifa, E.N., Martin, H., 1990. Major and trace element geochemistry of the Archaean Ntem plutonic complex (south Cameroon); petrogenesis and crustal evolution. Precambrian Res. 47 (1–2), 35–50.

Nzenti, J.P., Barbey, P., Macaudiere, J., Soba, D., 1988. Origin and evolution of the late Precambrian high-grade Yaounde gneisses (Cameroon). Precambrian Res. 38 (2), 91–109.

Rolin, P., 1992. Presence d'un chevauchement ductile majeur d'age panafricain dans la partie centrale de la Republique Centrafricaine: resultats preliminaires. C.R. Acad. Sci. Paris Ser. II 315, 467–470.

Samson, S.D., Alexander, E.C., 1987. Calibration of the interlaboratory $^{40}Ar–^{39}Ar$ dating standard, MMhb-1. Chem. Geol. Isotope Geosci. Sect. 66 (1–2), 27–34.

Tohver, E., D'Agrella-Filho, M.S., Trindade, R.I.F., 2006. Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambrian Res. 147 (3–4), 193–222.

Toteu, S.F., Van Schmus, W.R., Penaye, J., Nyobe, J.B., 1994. U–Pb and Sm–Nd evidence for Eburnian and Pan-African high-grade metamorphism in cratonic rocks of southern Cameroon. Precambrian Res. 67 (3–4), 321–347.

Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., Brito Neves, B.B., 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth Planetary Sci. Lett. 244 (1–2), 361–377.

Van Der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184 (1), 1–9.

Vaughan, A.P.M., Pankhurst, R.J., 2008. Tectonic overview of the West Gondwana margin. Gondwana Res. 13, 150–162, doi:10.1016/j.gr.2007.07.004.

Vicat, J.-P., Léger, J.M., Nsifa, E., Piguet, P., Nzenti, J.P., Tchameni, R., Pouclet, A., 1996. Distinction au sein du craton congolais du Sud-Ouest du Cameroun, de deux èpisodes dolèitiques initiant les cycles orogeniques eburneèen (Palèoprotèrozoique) et panafricain (Nèoprotèrozoique). C.R. Acad. Sci. Paris, Ser. II 323, 575–582.

Wilde, S.A., Youssef, K., 2000. Significance of SHRIMP U–Pb dating of the imperial porphyry and associated Dokhan Volcanics, Gebel Dokhan, north Eastern Desert, Egypt. J. Afr. Earth Sci. 31 (2), 403–413.